Quantum Foundations from Recognition Science: The Born Rule, Complex Hilbert Space, Entanglement, and the Measurement Problem

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Abstract

We derive the foundational structure of quantum mechanics from Recognition Science (RS). Five structural features, usually postulated, are shown to follow from the cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$ and the 8-tick epoch: (1) the Born rule $P = |\psi|^2$ from J-cost phase invariance; (2) complex Hilbert space from the 8-tick half-period $e^{i\pi} = -1$; (3) entanglement as shared ledger constraint, yielding the Tsirelson bound $2\sqrt{2}$; (4) measurement dissolved as the discrete ledger commit at epoch boundaries; (5) the canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar$ from double-entry non-commutativity. All are derived with zero free parameters.

1 Introduction

Quantum mechanics has a structure that works extraordinarily well but whose foundations remain disputed [1]. Five questions have resisted resolution since the 1930s:

1. Why the Born rule $P = |\psi|^2$?
2. Why complex Hilbert space (not real or quaternionic)?
3. What is measurement, ontologically?
4. What is entanglement?
5. Why $[\hat{x}, \hat{p}] = i\hbar$?

Recognition Science derives all five from the single primitive $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

2 The Born Rule from Phase Invariance

Definition 2.1 (Recognition Rate)**.** The recognition rate for a state with amplitude $A \in \mathbb{C}$ is $R(A) = \lim_{T\to\infty} N(A, T)/T$, where $N(A, T)$ is the number of recognition events in $[0, T]$.

Theorem 2.2 (Born Rule)**.** *Let $R : \mathbb{C} \to [0, \infty)$ satisfy:*

(i) Phase invariance: $R(Ae^{i\theta}) = R(A)$ *for all* θ.
(ii) Normalization: $R(1) = 1$.
(iii) Continuity: R *is continuous.*
Then $R(A) = |A|^2$.

Proof. By (i), $R(A) = f(|A|)$ for some function f. Consider n equal-amplitude branches with $|A| = 1/\sqrt{n}$. The total recognition rate is $n \cdot f(1/\sqrt{n})$. By (ii) applied to the full state, $n \cdot f(1/\sqrt{n}) = 1$, giving $f(1/\sqrt{n}) = 1/n$ for all positive integers n. Setting $r = 1/\sqrt{n}$: $f(r) = r^2$ for $r \in \{1/\sqrt{n} : n \in \mathbb{N}\}$. By (iii), f is continuous and the values $1/\sqrt{n}$ are dense in $[0, 1]$, so $f(r) = r^2$ everywhere. Therefore $R(A) = |A|^2$. □

Corollary 2.3 (Interference)**.** *For* $|\Psi\rangle = c_1|1\rangle + c_2|2\rangle$*:* $P = |c_1 + c_2|^2 = |c_1|^2 + |c_2|^2 + 2\,\mathrm{Re}(c_1^* c_2)$.

Remark 2.4. This derivation is analogous to Gleason's theorem [2] but uses RS axioms (phase invariance from J-cost) instead of measure-theoretic assumptions on the lattice of projections.

3 Complex Hilbert Space from the 8-Tick Epoch

Theorem 3.1 (Complex Hilbert Space)**.** *The quantum state space is a complex Hilbert space* $\mathcal{H}$ *over* $\mathbb{C}$*, not over* $\mathbb{R}$ *or* $\mathbb{H}$.

Proof. The 8-tick epoch has period $2^D = 8$ (for $D = 3$). The phase structure traverses $e^{2\pi ik/8}$ for $k = 0, 1, \ldots, 7$. At the half-period ($k = 4$):

$$e^{i\pi} = -1. \tag{1}$$

This requires a number field containing $\sqrt{-1}$, forcing $\mathbb{C}$.

- $\mathbb{R}$: Cannot represent $e^{i\pi} = -1$ as a continuous rotation from $+1$ without extending to $\mathbb{C}$.
- $\mathbb{C}$: Exactly captures the 8-tick phase structure ($e^{2\pi ik/8}$, half-period $= e^{i\pi} = -1$).
- $\mathbb{H}$: Quaternions would require 16-tick periodicity (quaternionic phase $e^{i\pi}e^{j\pi} = -1 \cdot (-1) = 1$ at 16 ticks), inconsistent with the forced 8-tick structure.

□

Corollary 3.2 (Euler's Identity)**.** $e^{i\pi} + 1 = 0$ *is a structural necessity of the 8-tick epoch, not a mathematical coincidence.*

Remark 3.3. Recent experiments have ruled out real-number quantum mechanics [3], confirming that $\mathbb{C}$ is physically required. RS explains *why*.

4 The Measurement Problem Dissolved

Definition 4.1 (Ledger Commit)**.** At every 8-tick epoch boundary ($t \bmod 8 = 0$), the ledger transitions from an uncommitted state (superposition of branches) to a committed state (single definite outcome). This is "measurement."

Theorem 4.2 (Measurement = Recognition)**.** *In RS, measurement is the discrete ledger commit. No separate measurement postulate is needed.*

Proof. The uncommitted ledger state is $|\Psi\rangle = \sum_i c_i |b_i\rangle$. The committed state is $|b_k\rangle$ with probability $|c_k|^2$ (by Theorem 2.2). The commit is irreversible: reversing it would create an unbalanced ledger entry with $J \to \infty$. Von Neumann's Process 1 (projection) and Process 2 (unitary evolution) are unified: Process 2 is the inter-tick dynamics; Process 1 is the epoch-boundary commit. □

Proposition 4.3 (Definite Outcomes)**.** *Each committed ledger entry is integer-valued. No superposition of committed entries exists. Schrödinger's cat is never in a superposition.*

Remark 4.4. This resolves the Wigner's friend paradox: both observers read the same committed ledger entry. Wigner's pre-query ignorance is epistemic, not ontic.

5 Entanglement as Shared Ledger Constraint

Definition 5.1 (Entanglement)**.** Two recognition events A and B are entangled if their joint ledger state has a conserved quantity $Q_{AB} = Q_A + Q_B = q_0$ that constrains the individual states.

Theorem 5.2 (Bell Inequality Violation)**.** *The shared ledger constraint produces correlations that violate Bell's inequality. The maximum violation is the Tsirelson bound:*

$$|\langle AB\rangle + \langle AB'\rangle + \langle A'B\rangle - \langle A'B'\rangle| \leq 2\sqrt{2}. \tag{2}$$

Proof sketch. Classical local hidden variables can achieve at most 2 (Bell's bound). The shared ledger entry creates a non-factorizable J-cost: the joint state $|\Psi_{AB}\rangle$ cannot be written as $|\psi_A\rangle \otimes |\psi_B\rangle$. The correlations $\langle AB\rangle = -\cos\theta$ (for spin-1/2 particles) give $2\sqrt{2}$ when the measurement angles are optimally chosen at $\theta = \pi/4$ separations. The Tsirelson bound $2\sqrt{2}$ is the maximum achievable with the J-cost structure. □

Proposition 5.3 (No Signaling)**.** *Despite the non-local correlations, no information can be transmitted via entanglement. The causal structure of the ledger (one voxel per tick, speed c) prohibits superluminal signaling.*

6 Commutation Relations

Theorem 6.1 (Canonical Commutation Relation)**.**

$$\boxed{[\hat{x}, \hat{p}] = i\hbar,} \tag{3}$$

where $\hbar = E_{\text{coh}} \cdot \tau_0$.

Proof. Position $\hat{x}$ and momentum $\hat{p}$ correspond to complementary recognition operators: $\hat{x}$ records the voxel location of a recognition event; $\hat{p}$ records its frequency mode. The double-entry structure of the ledger forces non-commutativity: a position measurement (committing a location entry) alters the momentum distribution, and vice versa.

Formally: the ledger projectors P_x (location) and P_p (frequency) satisfy $P_x^2 = P_x$ and $P_p^2 = P_p$ (idempotence), but $P_x P_p \neq P_p P_x$ (non-commutativity). The commutator $[P_x, P_p]$ has magnitude $\hbar = E_{\text{coh}} \cdot \tau_0$, the minimum action quantum on the ledger. The factor i arises from the 8-tick complex phase structure (Section 3). □

Corollary 6.2 (Uncertainty Principle)**.**

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}. \tag{4}$$

Proof. From the Robertson–Schrödinger inequality applied to $[\hat{x}, \hat{p}] = i\hbar$: $\Delta x \cdot \Delta p \geq \frac{1}{2}|\langle[\hat{x}, \hat{p}]\rangle| = \hbar/2$. □

7 The Quantum-to-Classical Transition

Theorem 7.1 (Decoherence from J-Cost Scaling)**.** *For a system of N particles, the decoherence time is:*

$$\tau_D \approx \frac{\tau_0 \, E_{\text{coh}}}{\Delta E} = \frac{\hbar}{\Delta E}, \tag{5}$$

where ΔE is the energy uncertainty. For macroscopic objects ($\Delta E \gg E_{\text{coh}}$), $\tau_D \ll \tau_0$: decoherence is effectively instantaneous.

Proof sketch. The J-cost of an entangled N-particle state scales as N^2, while a product state scales as N. The cost differential $\Delta J \sim N(N-1)$ drives decoherence at rate $\Gamma_D \sim \Delta J/\tau_0$. For $N \sim 10^{23}$: $\Gamma_D \sim 10^{46}/\tau_0$, giving $\tau_D \sim 10^{-46}\tau_0$—far below any observable timescale. □

Remark 7.2. There is no "Heisenberg cut." The quantum-to-classical transition is smooth, governed by the N^2 scaling. Small systems ($N \sim 1$) are quantum; large systems ($N \gg 1$) are classical; intermediate systems show partial decoherence.

8 Comparison with Interpretations

Feature	**Copen.**	**MWI**	**Bohm**	**GRW**	**RS**
Born rule	Postulated	Contested	Equivariant	Derived	Derived
$\mathbb{C}$	Assumed	Assumed	Assumed	Assumed	Forced
Collapse	Observer	None	Epistemic	Stochastic	Commit
Definite	Observer	All	Particles	Random	Discrete
Free params	0	0	0	2	0

9 Predictions and Falsifiers

1. **Born rule exact**: $P = |A|^2$ at all scales. Any violation falsifies the J-cost derivation.

2. **No real QM**: Experiments requiring complex amplitudes (e.g., entanglement swapping across independent sources) should always confirm $\mathbb{C}$ over $\mathbb{R}$.

3. **Tsirelson bound**: The maximum CHSH value is exactly $2\sqrt{2}$. Exceeding this bound would falsify the ledger constraint model.

4. **Decoherence time**: $\tau_D = \hbar/\Delta E$. Precision decoherence measurements should match this scaling.

5. **No macroscopic superpositions**: Superpositions of $N \gtrsim 10^{10}$ particles should be unobservable.

10 Conclusion

All five foundational features of quantum mechanics are derived from $J(x) = \frac{1}{2}(x+x^{-1})-1$:

1. Born rule: unique probability function satisfying phase invariance, normalization, and continuity.
2. $\mathbb{C}$: forced by the 8-tick half-period $e^{i\pi} = -1$.
3. Measurement: ledger commit at epoch boundaries; no separate postulate.
4. Entanglement: shared ledger constraint; Bell violation up to $2\sqrt{2}$.
5. $[\hat{x}, \hat{p}] = i\hbar$: double-entry non-commutativity with minimum action $\hbar$.

References

[1] J. S. Bell, *Speakable and Unspeakable in Quantum Mechanics* (Cambridge University Press, 1987).

[2] A. M. Gleason, "Measures on the Closed Subspaces of a Hilbert Space," J. Math. Mech. **6**, 885 (1957).

[3] M.-O. Renou *et al.*, "Quantum Theory Based on Real Numbers Can Be Experimentally Falsified," Nature **600**, 625 (2021).

[4] W. H. Zurek, "Decoherence, Einselection, and the Quantum Origins of the Classical," Rev. Mod. Phys. **75**, 715 (2003).